Tokenization Gap in BERT-Based Sentiment Analysis

Part 3: Exploring Tokenization Strategies on the IMDb Dataset

1. Introduction

WordPiece tokenization, the default mechanism used in BERT (Bidirectional Encoder Representations from Transformers), is a static subword segmentation algorithm trained on large general-purpose corpora. While effective for many NLP tasks, it may not be optimal for domain-specific or structurally diverse text such as movie reviews. This report investigates the Tokenization Gap in BERT-based sentiment analysis using the IMDb dataset.

The central research question is: does the choice of tokenization strategy significantly affect the downstream performance of a fine-tuned BERT model? We evaluate four tokenization approaches — WordPiece (baseline), Byte Pair Encoding (BPE), Character-Level, and a Hybrid Word + Character method — on a binary sentiment classification task, and further explore two extension techniques: Hybrid tokenization and Dynamic Token Merging.

2. Related Work

Tokenization is a foundational step in NLP pipelines, directly influencing model vocabulary coverage, sequence length, and ultimately performance. Three broad paradigms dominate modern NLP tokenization:

- WordPiece (Schuster & Nakamura, 2012; Devlin et al., 2019): Used natively by BERT, this algorithm splits words into subword units based on likelihood maximization over a training corpus. It balances vocabulary size with coverage of rare or compound words.
- Byte Pair Encoding / BPE (Sennrich et al., 2016): Originally developed for neural machine translation, BPE iteratively merges the most frequent character pairs to form subword units. It is data-driven and can be trained on task-specific corpora.
- Character-Level Models (Zhang et al., 2015): Represent text as individual characters, yielding a tiny vocabulary but very long sequences. These models can capture morphological patterns but suffer from information loss under fixed-length truncation.

Several studies highlight that tokenization mismatches can degrade BERT fine-tuning, especially for specialized domains. Hybrid strategies have been proposed as a compromise, leveraging word-level efficiency while using character-level fallback for out-of-vocabulary (OOV) tokens.

3. Methodology

3.1 Dataset

The IMDb Large Movie Review Dataset was used, comprising 25,000 training and 25,000 test reviews labeled for binary sentiment (positive/negative). To ensure tractable training times, 2,000 randomly shuffled training samples and 500 test samples were selected — with explicit shuffling (seed=42) applied before slicing to guarantee label balance.

- Train Set: 2,000 samples — ~1,000 positive, ~1,000 negative
- Test Set: 500 samples — ~250 positive, ~250 negative
- Maximum Sequence Length: 128 tokens (with truncation and padding)

3.2 Model

All experiments use bert-base-uncased (110M parameters, ~418 MB), fine-tuned for 2 epochs with a batch size of 16 (train) and 32 (eval), with FP16 mixed precision on GPU. The Hugging Face Trainer API was used consistently across all experiments, with eval_strategy='epoch'.

3.3 Tokenization Strategies

Four tokenization strategies were evaluated:

- WordPiece (Baseline): Direct use of BertTokenizer from bert-base-uncased. Vocab size ~30,000.
- BPE: A custom BPE tokenizer trained on 2,000 training texts using the HuggingFace tokenizers library, with vocab_size=10,000. Token sequences were then re-encoded through the BERT tokenizer for compatibility.
- Character-Level: Each character (up to first 300 characters per review) is tokenized individually and then passed through the BERT tokenizer. This yields a tiny vocabulary (~90 tokens) but extremely long sequences.
- Hybrid (Word + Char): An in-vocabulary word set is built from training data. Known words are passed as-is; unknown words are decomposed character by character. This provides OOV robustness with shorter average sequence lengths.

3.4 Extension Techniques

- Hybrid Tokenization: Acts as a middle ground between BPE and character-level, achieving a word vocab of ~20,000 from training data while preserving character-level fallback.
- Dynamic Token Merging: Adjacent short tokens (length ≤ 2) are merged pairwise. This reduces sequence length for character-level representations and partially recovers information lost due to MAX_LEN=128 truncation.

4. Experiments

4.1 Experimental Setup

All four tokenizers encoded train and test texts into fixed-length (128) input_ids, attention_masks, and token_type_ids. Each configuration was passed through the same fine-tune_bert() function, ensuring no variation in training hyperparameters. Accuracy and weighted F1-score were computed using scikit-learn on the held-out test set after 2 epochs.

4.2 Dynamic Token Merging Demo

Example on the sentence: "The movie was absolutely brilliant!"

- WordPiece tokens: ['the', 'movie', 'was', 'absolutely', 'brilliant', '!']
- BPE tokens: ['The', 'movie', 'was', 'absolutely', 'brilliant', '!']
- Char tokens: ['T', 'h', 'e', ' ', 'm', 'o', 'v', 'i', 'e', ' ', 'w', 'a', 's', ...]
- After Dyn Merge: ['Th', 'e ', 'mo', 'vi', 'e ', 'wa', 's ', 'ab', 'so', 'lu', 'te', 'ly', ' b', 'ri', 'll', 'ia', 'nt', '!']

Dynamic merging reduced 37 character tokens to 18 merged tokens, nearly halving sequence length and allowing more of the review text to fit within the 128-token window.

4.3 Tokenization Statistics (100 Samples)

WordPiece	347.6	5320	BERT default
BPE	351.0	5129	Data-driven subword
Character-Level	1479.3	90	Very long sequences
Hybrid	67.7	1895	Word + char fallback

Character-Level tokenization produces sequences nearly 4.3x longer than WordPiece (1479.3 vs 347.6 avg tokens), meaning only a small fraction of each review is captured within 128 tokens. Hybrid tokenization achieved the shortest average sequence length (67.7), benefiting from treating known words as single tokens.

5. Results

5.1 Final Results Table

WordPiece (Baseline)	0.8520	0.8518	45.5s	418
BPE	0.8620	0.8619	44.7s	418
Character-Level	0.5080	0.3423	46.7s	418
Hybrid (Word + Char)	0.8260	0.8258	45.3s	418

Best F1: BPE (F1 = 0.8619) | Fastest: BPE (44.7s)

BPE achieved the highest accuracy (86.20%) and F1 score (0.8619), outperforming the WordPiece baseline by approximately 1 percentage point in both metrics. The Hybrid tokenizer performed comparably to WordPiece (Accuracy: 82.60%, F1: 0.8258), while Character-Level tokenization exhibited severe degradation with an F1 of only 0.3423 — barely above random chance for binary classification.

6. Analysis

6.1 WordPiece (Baseline)

WordPiece is purpose-built for BERT and serves as a strong baseline. With an average of 347.6 tokens per review (well within the 128-token window after truncation) and a task-relevant vocabulary, it achieves 85.20% accuracy. The slight underperformance relative to BPE is likely because WordPiece was trained on Wikipedia/BookCorpus rather than movie review text, leading to suboptimal subword boundaries for domain-specific vocabulary.

6.2 BPE — Best Performer

BPE outperformed all methods, achieving the highest F1 (0.8619) and the fastest training time (44.7s). Training BPE directly on IMDb reviews allows it to learn subword boundaries specific to the sentiment domain — words like 'cinematography', 'predictable', or 'heartwarming' are common in movie reviews and BPE handles them more naturally. Additionally, BPE's better OOV handling compared to WordPiece (which relies on [UNK]) contributes to improved representation quality.

6.3 Character-Level — Severe Degradation

Character-Level tokenization produced the worst results by a large margin (F1: 0.3423, Accuracy: 50.80%). The primary reason is the information bottleneck imposed by MAX_LEN=128: with an average of 1479.3 character tokens per sample, only the first ~128 characters (~25 words) of each review are seen by the model. Sentiment information is distributed across the entire review, and the early words alone are often insufficient for reliable classification. Furthermore, BERT's positional embeddings and attention patterns are not optimized for character-level granularity.

6.4 Hybrid Tokenization — Practical Middle Ground

The Hybrid tokenizer achieved 82.60% accuracy and 0.8258 F1, positioned between WordPiece and BPE in performance. Its average token count of 67.7 per review is the lowest across all methods, meaning more text fits within the 128-token window — yet performance is slightly lower than WordPiece. This suggests that while sequence efficiency improves, the word-level tokens from training vocabulary may not align perfectly with BERT's pre-trained WordPiece representations, causing embedding mismatches. This hybrid approach, however, shows strong potential when applied to domains with heavy morphological variation or out-of-vocabulary terms.

6.5 Dynamic Token Merging

Dynamic Token Merging demonstrates a viable approach to mitigate the sequence-length problem of character-level models. By merging adjacent short tokens, it reduces a 37-character sequence to 18 merged tokens for the test example — theoretically allowing more of the review to fit in the 128-token window. Applied at scale, this could partially recover the information loss responsible for Character-Level's poor performance, though its integration into a full training pipeline was demonstrated as a proof-of-concept in this study.

6.6 Summary of Key Insights

- WordPiece provides a reliable, well-calibrated baseline that is hard to beat without domain-specific adaptation.
- BPE trained on in-domain data consistently outperforms WordPiece, confirming the value of domain-adapted tokenization.
- Character-Level tokenization is fundamentally incompatible with fixed-length BERT models at moderate MAX_LEN values; it requires either dynamic length handling or dedicated architectures.
- Hybrid tokenization is a practically effective strategy offering OOV robustness and sequence efficiency, with room for further optimization.
- Dynamic Token Merging shows promise as a post-processing step to recover information lost in character-level truncation.